

Arindam Shiva Tripathi

+1 (227) 249-4304 | aritrip@umd.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Maryland (UMD)

College Park, MD, USA

Master of Science in Human-Computer Interaction (HCIM) — GPA 4.0

Aug 2025 – May 2027

- M.S. Thesis (in progress), building on Reframed; advised by Dr. Ming Lin and Dr. Beth Bonsignore

National University of Singapore (NUS)

Singapore

Bachelor of Engineering in Computer Engineering, Minor in Business Management

Aug 2019 – Jul 2023

- Concentration in Interactive Digital Media; graduated with honors

EXPERIENCE

LIGHT Helmets

Carlsbad, CA (Remote)

3D Scanning & Fitment Software Intern

May – Aug 2026

- Sole developer of an end-to-end Python scan-to-fitment pipeline for two helmet families, turning 3D head and helmet-interior scans into per-station padding recommendations
- Built an Android head-scanning app in Kotlin using ARCore's Depth API to capture 3D head geometry
- Automated helmet placement on 3D head scans by searching 52k positions, cutting head-helmet collisions from 141 points to 0 after a standard optimizer got stuck

L'Oréal

Singapore

Data Analytics Specialist (Full-Time)

Jul 2023 – Aug 2025

- Built ML-backed planning tools (Python, SQL, React) used by 200+ APAC planners, improving efficiency by 24%
- Implemented data validation rules across shared database views to standardize reporting workflows
- Increased system adoption by 40% through iterative feature refinement and usage monitoring

Enterprise Architecture Intern

May 2022 – Sep 2022

- Optimized backend data structures, reducing database size by 30% and improving dashboard load times by 20%

SP Digital

Singapore

Full Stack Developer Intern

Jun 2021 – Dec 2021

- Shipped production React/JavaScript features for utility workflows in the SP Digital app (1.2M+ downloads)
- Debugged state handling and API request flows in an internal Golang portal, fixing production transaction bugs

PROJECTS

SnaPT — ChatGPT Context-Management Extension

Apr 2026

JavaScript, Chrome Extension APIs

- Shipped a Chrome Web Store extension that cut a long ChatGPT thread's load time from 22s to 10s (2.2× faster) by trimming active context while keeping older turns recoverable
- Reverse-engineered ChatGPT's conversation payload to trim it before render, after DOM-only hiding failed to help
- Redesigned the data model around grouped turns (a user message plus its replies) after per-message counting confused users, and shipped privacy-first with single-site permissions, local-only storage, and no remote code

Aura Desk — Ambient AR Productivity Surface

Jun 2026 – Present

Kotlin, Jetpack XR SDK, Android XR, On-Device AI

- Built a native Kotlin app on the pre-release Jetpack XR SDK for unreleased Xreal Project Aura smart glasses
- Implemented spatially anchored widgets (Pomodoro, tasks, calendar, music) that pin to a physical desk
- Integrated on-device AI with voice commands so natural-language requests run fully offline

Reframed — Real-Time Computer Vision for Passthrough MR

Mar 2026 – Present

C#, Unity, OpenCV

- Built a real-time pipeline that recognizes physical artworks through a headset's passthrough camera via ORB feature matching and homography, estimating 6-DoF pose with solvePnP to anchor content
- Demoed live to 50+ visitors over two days at UMD's NextNOW Fest

SKILLS

Languages: Python, Kotlin, JavaScript, TypeScript, C#, C/C++, Go, SQL

Frameworks & Tools: React, Android (Jetpack XR SDK, ARCore), Unity, OpenCV, Docker

Data & ML: NumPy, pandas, Open3D, trimesh, PyTorch, scikit-learn, Vertex AI (GCP)

Credentials: Google Trusted Developer, Android XR (2026); Android Dev Summit invitee; Google Cloud Digital Leader